9/7 Group Meeting

From Cotangent Bundles to Lagrangian Fibrations

§ Intro (IHS)

K3 $\approx$ $\exists$ nowhere-degenerate holomorphic 2-form.

Natural Question

$\boxed{\text{their higher-dimensional analogues?}}$

$\rightsquigarrow$ Irreducible Holomorphic Symplectic manifold

Def An IHS mfd is a compact Kähler manifold such that
↳ In particular, smooth?

- $\pi_1(X) = 0$
- $H^0(X, \Omega_X^2) = \mathbb{C}\sigma$

for an everywhere non-degenerate holomorphic 2-form

Rmk 1 [Non-degenerate]

For every $x \in X$

$\sigma_x : T_x X \times T_x X \longrightarrow \mathbb{C}$ skew symm bilinear form

s.t $\sigma_x(v, w) = 0$ for all $w \in T_x X \Rightarrow v = 0$.

i.e $\boxed{\begin{array}{ll} TX \xrightarrow{\sim} T^*X & \text{bundle map} \\ v \longmapsto \iota_v \sigma = \sigma(v, -) & \end{array}}$

or $T_X \simeq \Omega_X^1$ as a sheaf.

moreover

$\Leftrightarrow \sigma^{\wedge n} \neq 0$ everywhere

$(\Leftrightarrow \sigma^{\wedge n} \in H^0(X, \Omega_X^{2n}) = H^0(X, K_X))$

$\Rightarrow \dim X = 2n, \quad K_X \simeq \mathcal{O}_X$

Rmk 2 [Real v.s. Holomorphic]

Real M^{2n} : sm real mfd w/ closed non-deg 2-form $\omega \in \Omega^2(M, \mathbb{R})$

hol X^{2n} : complex mfd w/ hol non-deg 2-form $\sigma \in H^0(X, \Omega_X^2)$.

Locally, the two theories look very similar.

Darboux coordinate $\omega = \sum_i dq_i \wedge dp_i$

$\sigma = \sum_i dz_i \wedge dw_i$

Rmk 3.

"Irreducible" $\longrightarrow$ $\pi_1(X) = 0$
excludes torus factors.

$\longrightarrow$ $h^{2,0}(X) = 0$
excludes products of several HS factors.

e.g. abelian surf : $h^{2,0} = 1$ but not simply conn.
$K3 \times K3$: simply conn but $h^{2,0} = 2$.

$\rightsquigarrow$ they ensure "X is an indecomp. building block in hol. symp. geom.

We know surprisingly few examples.

- $\dim X = 2$: K3 surf.
- Higher dim

Def. type	dim
$K3^{[n]}$	$2n$
Kum^n	$2n$
OG6	6
OG10	10

$S^{[n]}$ = Hilbert sch of points on a K3
$= \{0\text{-dim subsch of } S \text{ of length} = n\}$

$\dim S^{[n]} = 2n$

[principle]
higher dim IHS often arise naturally as moduli spaces

How can we construct them ? $\boxed{\text{Key idea}}$ (deformation + duality)

§ Moduli-theoretic constructions of IHS.

One of the most important sources : Moduli theory.

$$S^{[n]} = \{ I_Z : \text{ideal sheaf} \} = M_v(1,0,1-n)$$

Q Why should a moduli space of sheaves on a K3 surf carry a symp form?

S : K3, M : 'appropriate' moduli space of stable sheaves on K3

$$[E] \in M \rightsquigarrow \boxed{T_{[E]} M \simeq \mathrm{Ext}^1(E,E)}$$

infinitesimal def

Yoneda product + tr. map

$$\mathrm{Ext}^1(E,E) \times \mathrm{Ext}^1(E,E) \longrightarrow \mathrm{Ext}^2(E,E) \xrightarrow{tr} H^2(S,\mathcal{O}_S) \underset{K3}{\simeq} \mathbb{C}$$

$\overset{\text{comp.}}{\Longrightarrow}$ bilinear form $\mathrm{Ext}^1(E,E) \times \mathrm{Ext}^1(E,E) \longrightarrow \mathbb{C}$

$$(v, w) \longmapsto tr(v \circ w)$$

by graded symmetry

$\rightsquigarrow tr(v \circ w) = -tr(w \circ v)$ $\boxed{\text{alternating}}$

by Serre duality $\mathrm{Ext}^i(E,F) \times \mathrm{Ext}^{2-i}(F, E \otimes K_S) \longrightarrow \mathbb{C}$ perfect

$i = 1$, $E = F$, $K_S \cong \mathcal{O}_S$ $\rightsquigarrow \mathrm{Ext}^1(E,E) \times \mathrm{Ext}^1(E,E) \longrightarrow \mathbb{C}$ perfect.

$\boxed{\text{non-degeneracy}}$

$$\boxed{\begin{array}{l}\text{Deformation theory + Serre duality} \\ \quad \Rightarrow \text{symplectic form}\end{array}}$$

The essential feature is not only the K3 surf itself, but the categorical property that the Serre functor is the shift by 2.

K3 category : $S_{\mathcal{C}} \simeq [2]$
(2-CY)

⚠ Such categories also arise from varieties that are not themselves holomorphic symplectic.

e.g. Kuznetsov component of a cubic fourfold / Gushel-Mukai varieties

K3 category + moduli of stable objects
$\Rightarrow$ holomorphic symp objects.

This is one of the main modern approaches to construct IHS.

Today focus on much more geometric viewpoint!

The most elementary model of symplectic geom.

phase space : T^*B.

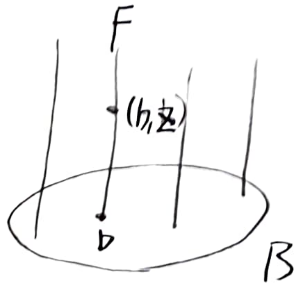
§ Cotangent bundles and Lagrangian fibrations.

Recall Real symplectic geom. : phase space

$T^*B \to B$ w/ local coordinate (q_i, p_i)
position momentum.

and canonical symp form (closed non-deg 2form $\in \Omega^2(M, \mathbb{R})$)

$w = \sum dq_i \wedge dp_i$ pairs the base direction $v \in T_bB$ with the fiber direction $\xi \in T_zF \simeq T_b^*B$



$T_b^*B \simeq T_zF$ -vect sp.

$w((v_1,\xi_1),(v_2,\xi_2)) = \xi_2(v_1) - \xi_1(v_2)$

Fiber T_b^*B are half dim & Lagrangian $(v=0)$

same picture exists in the hol. setting

$$\sigma = \sum dz_i \wedge dw_i$$

Now consider a Lagrangian fibration

$$\boxed{\pi : X^{2n} \longrightarrow B^n \quad \text{on a IHS} \quad \text{w/ } \sigma|_{\text{gen fiber}} = 0.}$$

For sm fiber F_b

$$0 \to T_x F_b \to T_x X \to T_b B \to 0$$

Since F_b is Lagrangian, it induces a perf pairing.

$$T_x F_b \times T_b B \to \mathbb{C}$$

$$\rightsquigarrow \boxed{T_x F_b \simeq T_b^* B}$$

So infinitesimally, the fiber directions

$$\|$$

the cotangent directions.

However $T_b^* B \simeq \mathbb{C}^n$ is noncpt,

while the fibers of a compact Lag fib are compact complex tori.

$\rightsquigarrow$ Quotient the cotangent space by a lattice

$$T_b^* B \longrightarrow T_b^* B / \Lambda_b. \quad \text{"rolled up"}$$

At least locally over the smooth locus

Lagrangian fib $\doteqdot \{F_b \simeq T_b^* B / \Lambda_b\}_{b \in B}$. "heuristic"

§ Beauville–Mukai system (Jacobians as compactified cotangent direc.)

Above picture becomes concrete in BM system

S : K3

C : genus g curve $|C| \cong \mathbb{P}^g$ $\qquad K_C = (K_S \oplus N_{C/S})|_C$

$$T_{[C]}|C| \underset{\text{Emb def}}{\cong} H^0(C, N_{C/S}) \overset{\text{adjunction}}{=} H^0(C, K_C)$$

$$\Rightarrow T^*_{[C]}|C| \cong H^0(C, K_C)^*$$

But Jacobian is precisely

$$J(C) \cong H^0(C, K_C)^* / H_1(C, \mathbb{Z})$$

$$\Rightarrow \boxed{J(C) \cong T^*_{[C]}|C| / \Lambda_C} \quad \text{so } J(\mathcal{C}/|C|) \to |C|$$

"cotangent dir/lattice" $\qquad$ Lag fib.

This is the geometric reason why Jacobians appear so naturally in hol symp Lag. fib.!

Rem $X \sim T^*B/\Lambda$: only a local model over a smooth locus of the fibration.

$\exists$ global issues

- $\{\Lambda_b \subset T^*_b B\}$ must vary consistently with $b \in B$.
  $\rightsquigarrow$ local system w/ nontrivial monodromy
- Lag fib need not admit a global section
  $\rightsquigarrow$ fibers $\cong$ torsors under the corresp ab var.
- fiber degeneration $\rightsquigarrow$ replace by suitable compt'n.

Laza - Saccà - Voisin ('17)

$Y \subset \mathbb{P}^5$ sm cubic 4-fold.

$(\mathbb{P}^5)^\vee$: set of hyper plane

$\rightsquigarrow Y_H := Y \cap H \subset \mathbb{P}^4$: sm cubic 3-fold

for $H \in (\mathbb{P}^5)^\vee$

$\rightsquigarrow$ Int. Jac $J(Y_H) = H^3(Y_H, \mathbb{C}) / F^2 H^3(Y_H, \mathbb{C}) + H^3(Y_H, \mathbb{Z})$

is ab 5-fold
(non-trivial)

$\xrightarrow[\text{Donagi-Markman}]{} \mathcal{J}_U = \bigsqcup_{H \in U} J(Y_H)$ $U \subset (\mathbb{P}^5)^\vee$: sm hyperplane section locus

10-dim w/ hol symp form.

$\xrightarrow[\text{LSV}]{} \overline{\mathcal{J}}$: compactification

s.t $\overline{\mathcal{J}} \to (\mathbb{P}^5)^\vee$: Lag fib.

$\overline{\mathcal{J}}$: IHS of OG10-type

Hodge theory gives

$H^3(Y_H, \mathbb{C}) = H^{2,1}(Y_H) \oplus H^{1,2}(Y_H)$

$J(Y_H) = H^3(Y_H, \mathbb{C}) / (F^2 H^3(Y_H, \mathbb{C}) + H^3(Y_H, \mathbb{Z}))$

$\Rightarrow T_0 J(Y_H) = H^3(Y_H, \mathbb{C}) / \underbrace{F^2 H^3(Y_H, \mathbb{C})}_{\cancel{H^{3,0}()} \oplus H^{2,1}(Y_H, \mathbb{C})} \cong \boxed{H^{1,2}(Y_H)}$

On the other hand

$\boxed{T_H U \simeq H^{2,1}(Y_H)}$

dual (Hodge pairing)

§ From K3 to Ab surf

So far, $\boxed{\text{Cost. from K3}}$

$\boxed{\text{from Ab surf}}$

resulting moduli sp still retain the abelian directions such as translation and tensoring by line bundle

⟶ admit non-triv Alb map

⟶ we take an appropriate fiber of Alb map

moduli setting ⇝ gen Kum type

OG10 ⟶ OG6

BM ⟶ Debarre system.